Introduction to Composites Material

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Ceramics Matrix composites

Functions of the Matrix Material (Primary Phase)

- It takes the load and transfer it to the reinforcement

- It binds or holds the reinforcement phase and protects the same from external damage (either chemical and mechanical)

- Matrix also separates the individual fibers and prevents brittle cracks from passing completely across the section of the composite

Ceramic Matrix composites (CMCs)

Ceramic matrix composites (CMCs) are a special type of composite material in which both the reinforcement (refractory fibers) and matrix material are ceramics.

In some cases the same kind of ceramic is used for both parts of the structure, and additional secondary fibers may also be included.

Because of this, CMCs are considered a subgroup of both composite materials and ceramics.

Ceramics are widely used matrix materials for composites because of

- ✓High melting points
- ✓High compressive strength
- ✓Good Strength at elevated temperature
- ✓Excellent resistance to oxidation

However, ceramics are lacking

1. Poor tensile strength
2. Poor Impact strength
3. Thermal shock resistance

These limitation can be overcome by reinforcing ceramics matrices by suitable fiber materials

Common ceramic matrices material

1. Glass ceramics i.e., Lithium Alumino-silicate
2. Oxides i.e., Alumina and mullite
3. Silicon nitride
4. Silicon Carbide
5. Carbon, C

Need of CMCs

- Fracture toughness values for ceramic materials are low (1- 5 MPa).

- This can improve significantly by the development of a new generation of ceramic-matrix composites (CMCs)— particulates, fibres, or whiskers of one ceramic material that have been embedded into a matrix of another ceramic.
- Ceramic-matrix composite materials have extended fracture toughness to between about 6 and 20 MPa

In essence, this improvement in the fracture properties results from interactions between advancing cracks and dispersed phase particles. Crack initiation normally occurs with the matrix phase, whereas crack propagation is impeded or hindered by the particles, fibers, or whiskers.

Several techniques are utilized to retard crack propagation, which are discussed as follows.

Among them one particularly interesting and promising toughening technique employs a phase transformation to arrest the propagation of cracks and is termed transformation toughening

- Small particles of partially stabilized zirconia are dispersed within the matrix material, often Al_2O_3 or ZrO_2 itself.

- Typically, CaO, MgO, Y_2O_3 , and CeO are used as stabilizers. Partial stabilization allows retention of the metastable tetragonal phase at ambient conditions rather than the stable monoclinic phase; these two phases are noted on the ZrO_2–$ZrCaO_3$ phase diagram.

- The stress field in front of a propagating crack causes these metastable retained tetragonal particles to undergo transformation to the stable monoclinic phase.

- Accompanying this transformation is a slight particle volume increase, and the net result is that compressive stresses are established on the crack surfaces near the crack tip that tend to pinch the crack shut, thereby arresting its growth.

- This process is demonstrated schematically in below Fig

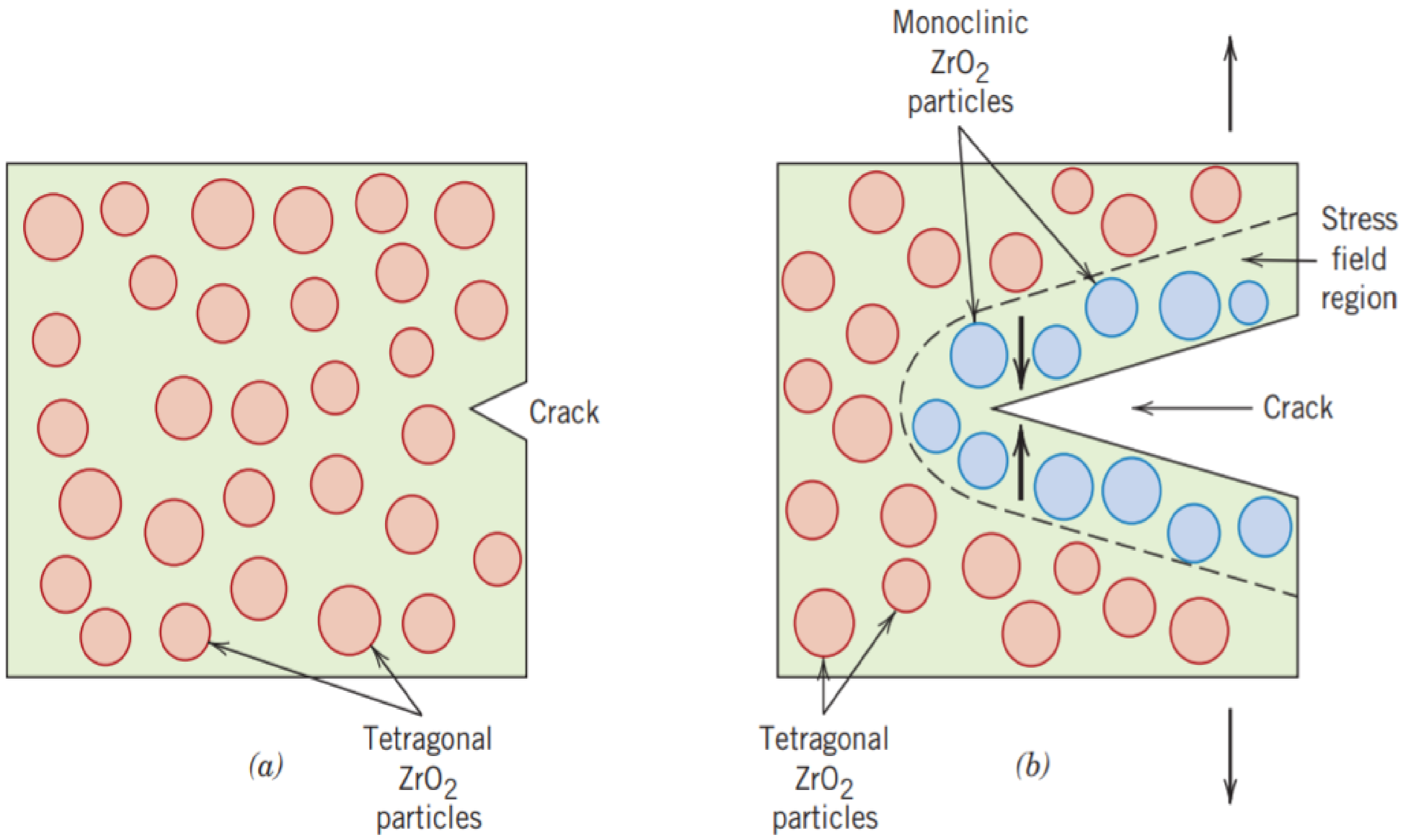


Fig. 1 Schematic demonstration of transformation toughening. (a) A crack prior to inducement of the ZrO2 particle phase transformation. (b) Crack arrestment due to the stress induced phase transformation.

Other recently developed toughening techniques involve the utilization of ceramic whiskers, often SiC or Si_3N_4.

These whiskers may inhibit crack propagation by

(1) Deflecting crack tips
(2) Forming bridges across crack faces
(3) Absorbing energy during pull-out as the whiskers deboned from the matrix
(4) Causing a redistribution of stresses in regions adjacent to the crack tips.

In general, increasing fiber content improves strength and fracture toughness; this is demonstrated in Table 16.10 for SiC whisker-reinforced alumina.

Table 1 Room Temperature fracture strengths and fracture toughness for various SiC whiskers content in Al2O3

Whisker Content (vol%)	*Fracture Strength (MPa)*	*Fracture Toughness ($MPa\sqrt{m}$)*
0	—	4.5
10	455 ± 55	7.1
20	655 ± 135	7.5–9.0
40	850 ± 130	6.0

Source: Adapted from *Engineered Materials Handbook*, Vol. 1, *Composites*, C. A. Dostal (Senior Editor), ASM International, Materials Park, OH, 1987.

Furthermore, there is a considerable reduction in the scatter of fracture strengths for whisker-reinforced ceramics relative to their unreinforced counterparts.

In addition, these CMCs exhibit improved high-temperature creep behaviour and resistance to thermal shock (i.e., failure resulting from sudden changes in temperature).

Ceramic-matrix composites may be fabricated using

1. Hot pressing
2. Hot isostatic pressing
3. Liquid phase sintering techniques.

Relative to applications, SiC whisker-reinforced alumina are being utilized as cutting tool inserts for machining hard metal alloys; tool lives for these materials are greater than for cemented carbides

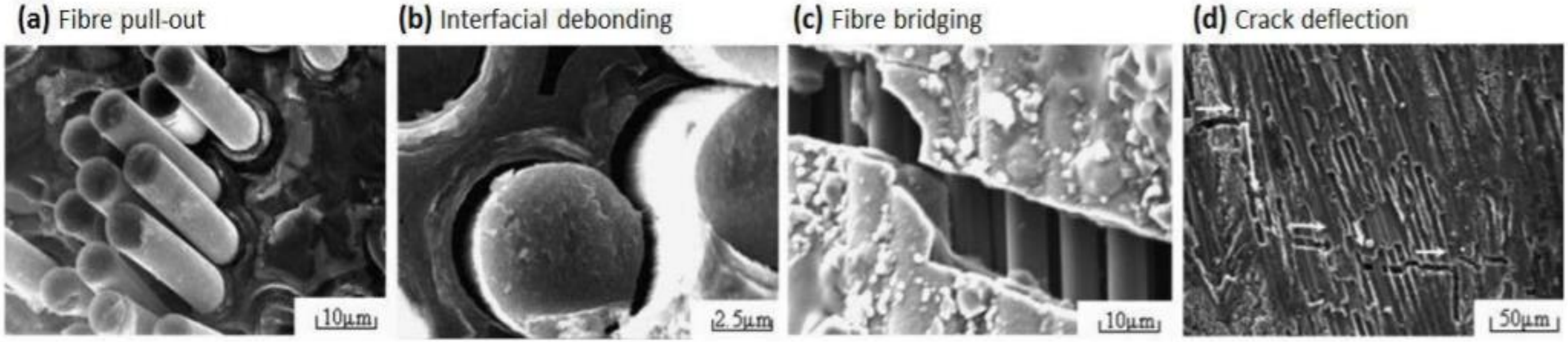
Ceramic Matrix Composites Manufacturing and Applications in context to Automotive Industry

Fiber Reinforced Ceramics

Fibre reinforcements can be used to improve the toughness of a material

High Temp in processing (and service) of CMC components

- Temperature resistance
- Chemical compatibility
- Thermal expansion mismatch



S. Fan, et al., Compos. Sci. Technol. 67, 2390 (2007).

Processing and Manufacture

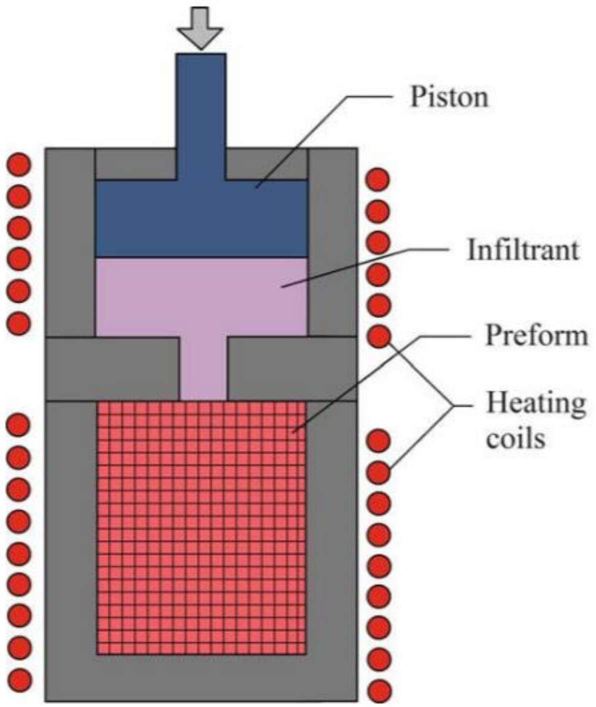
Conv. Ceramic Consolidation	Cold-Pressing and Sintering
Prepreg form	Slurry Impregnation and Hot-Pressing
	Melt Infiltration
	Sol-Gel Infiltration
	Polymer Infiltration and Pyrolysis (PIP)
Porous Preform Infiltration	Reactive Liquid Infiltration
	Directed Oxidation/Nitridation
	Reaction Bonding
	Chemical Vapor Infiltration (CVI)

Processing and Manufacture

Non-Reactive Liquid Infiltration

Melt Infiltration

1. Single step
2. High density
3. High melting temperatures
4. High viscosities



K. K. Chawla, in ***Compos. Mater.***, 3rd ed. Springer New York, New York, NY, 2012

Processing and Manufacture

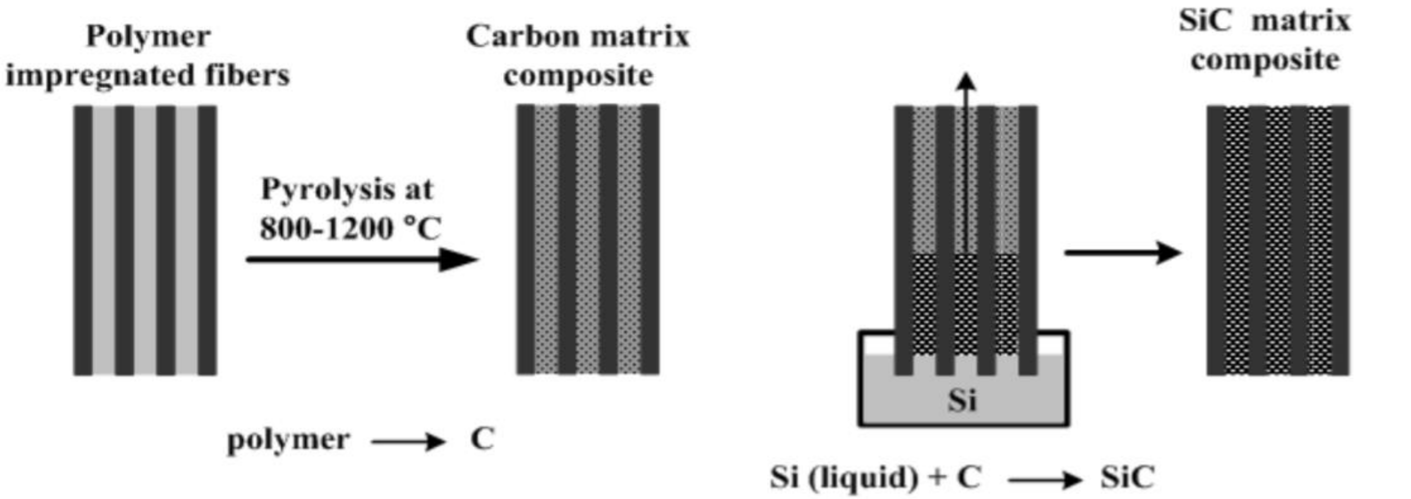
Reactive Liquid Infiltration

Liquid silicon infiltration (LSI)

First developed in late 1980s

Infiltration of C green-body with molten Si

$$C(s) + Si(l) \rightarrow SiC(s)$$



D. Kopeliovich, available at http://www.substech.com/ (Accessed 11/11/2014)

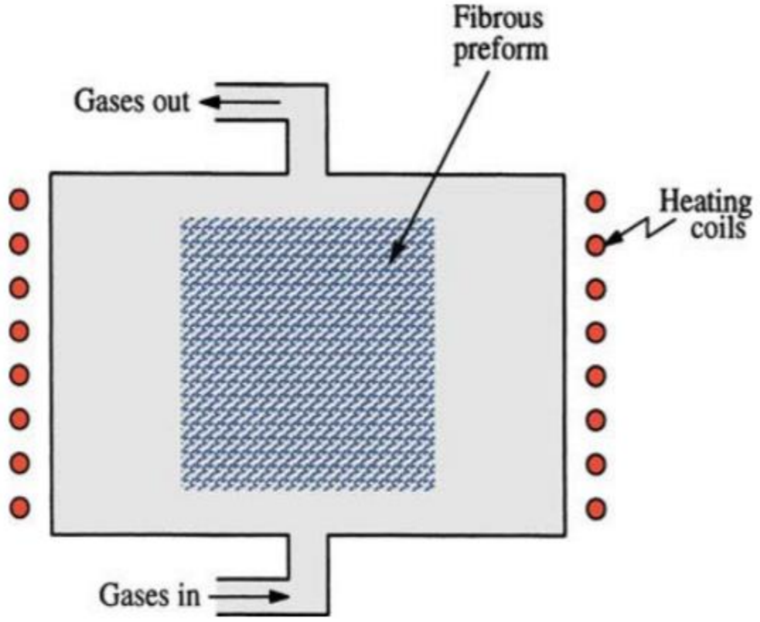
Processing and Manufacture

Gas Infiltration

Chemical Vapour Impregnation (CVI)

Ceramic Matrix	Precursors
SiC	CH_3SiCl_3
Si_3N_4	$SiCl_4 + NH_3$
Al_2O_3	$AlCl_3 + CO_2$
ZrO_2	$ZrCl_4 + CO_2$
TiB_2	$TiCl_4 + BCl_3$

- Almost any ceramic can be formed
- Near-net shape
- Slow process (diffusion)
- High cost



K. K. Chawla, in ***Compos. Mater.***, 3rd ed. Springer New York, New York, NY, 2012

Automotive Ind.- Braking systems

Ceramic composite brakes: C/SiC

1. High braking performance
2. Low weight (2.4 g/cm3)
3. Low wear rate
4. Operating temperatures 1,400°C

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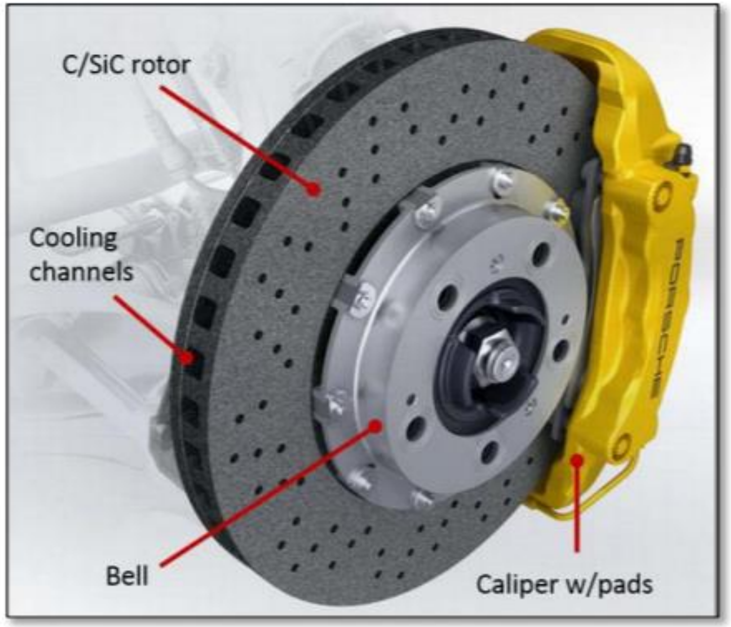


Porsche Cars Great Britain Ltd., available at http://www.porsche.com/uk/ (Accessed 15 Oct 2014).

First studied in 1990s, available in 2000s

1. Mercedes CL 55 AMG F1 Lim. Ed. (2000)
2. Porsche 911 GT2 (2001) (PCCB)

50,000-70,000 CMC brake discs manufactured in 2006

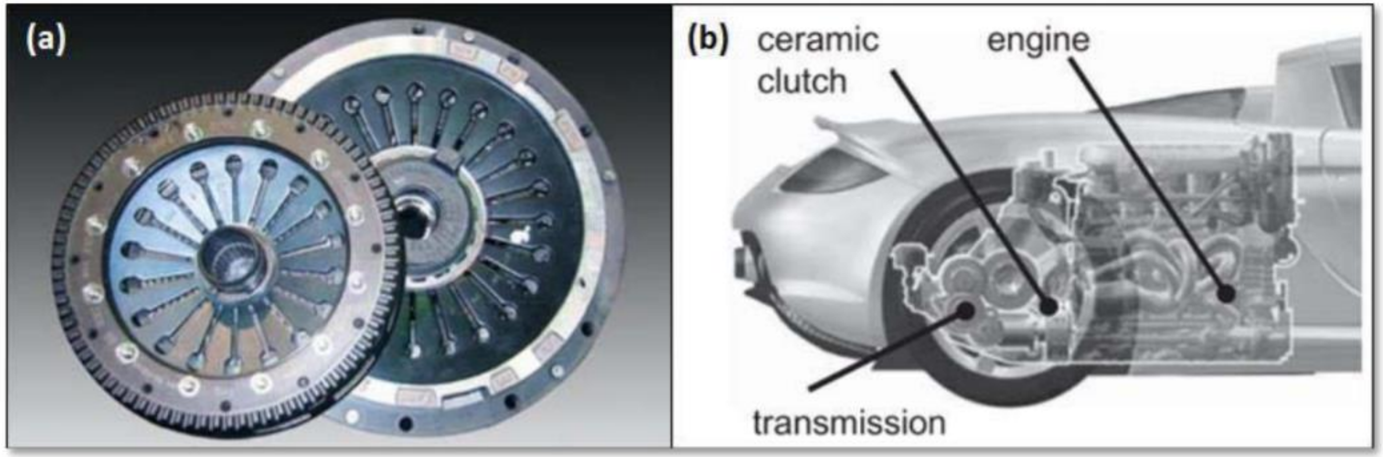
– SICOM™, BREMBO™, etc.

High Cost

Automotive Ind.- Clutches

Porsche Ceramic Composite Clutch (PCCC)

1. Specially designed for the Carrera GT
2. Siliconized carbon fibre fabrics
3. 169 mm- diameter, 3.5 kg
4. One tenth of mass moment of inertia
5. Lower transmission and engine mounting →Lower centre of gravity High cost



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FOR LISTENING

ANY QUESTIONS ??